

OC IT Projects — Chapter 10

Network Security — Review Sheet + Full Quiz

Based on the Chapter 10 PowerPoint content.

1. Key Points

- CIA Triad: Confidentiality, Integrity, Availability.
- Confidentiality: protects information from unauthorized access or disclosure.
- Integrity: protects data from unauthorized modification; hashing is presented under Integrity.
- Availability: keeps systems and information accessible to authorized users.
- Symmetric Encryption: same key for encryption and decryption.
- Asymmetric Encryption: uses a pair of keys.
- Hashing: used to help verify data integrity.
- Attacks: Man-in-the-Middle, Session Hijacking, DoS, DDoS, TCP SYN Flood, Smurf/ICMP Flood, Insider Threats, Logical Bomb, Phishing, Ransomware, VLAN Hopping.
- Security controls: Physical Controls, User Training, Vulnerability Scanners, Patching, Honey Pots/Honey Nets.
- Zenmap: associated in the training with scanning targets and identifying information such as open ports and OS information.
- Nessus: associated in the training with host scanning, including patching/update status.
- AUP: defines allowed and prohibited use of company resources.
- Standards: approved requirements. Guidelines: guidance. Procedures: specific steps.
- MFA factors: Something You Know, Something You Have, Something You Are, Something You Do, Somewhere You Are.

2. Your Mistakes in the Quiz

- Q13: TCP SYN Flood = large numbers of SYN requests targeting TCP connection resources.
- Q23: Patching addresses security weaknesses/vulnerabilities through updates.
- Q29: Fingerprint = Something You Are.
- Q30: Unauthorized file modification = Integrity; the chapter associates Hashing with Integrity.

3. Complete Quiz — All 30 Questions

Q1. Which three concepts make up the CIA Triad?

- A) Confidentiality, Integrity, Availability
- B) Control, Inspection, Access
- C) Confidentiality, Inspection, Authentication
- D) Control, Integrity, Authentication

Q2. Which CIA principle is primarily concerned with preventing unauthorized people from viewing information?

- A) Availability
- B) Confidentiality
- C) Integrity
- D) Authentication

Q3. Which CIA principle ensures that data has not been improperly modified?

- A) Availability
- B) Confidentiality
- C) Integrity
- D) Authorization

Q4. A company's website is online, but unauthorized users cannot access its confidential customer records. Which security goal is being demonstrated?

- A) Availability
- B) Integrity
- C) Confidentiality
- D) Load balancing

Q5. Which type of encryption uses the same key for encryption and decryption?

- A) Symmetric encryption
- B) Asymmetric encryption
- C) Hashing
- D) Multifactor encryption

Q6. Which statement best describes asymmetric encryption?

- A) It uses no keys
- B) It uses the same key for both operations
- C) It uses a pair of keys
- D) It is only used for hashing

Q7. Hashing is primarily associated with which CIA goal in this chapter?

- A) Availability
- B) Integrity
- C) Confidentiality
- D) Authentication

Q8. What is the primary purpose of hashing in the context presented?

- A) To verify that data has not been altered
- B) To make a network faster
- C) To provide physical security
- D) To prevent every type of attack

Q9. An attacker secretly positions themselves between two communicating parties and intercepts their communication. What attack does this describe?

- A) Logical bomb
- B) Ransomware
- C) Man-in-the-Middle
- D) VLAN hopping

Q10. An attacker takes over an already-established communication session between a user and a system. Which attack is this?

- A) Session hijacking
- B) Phishing
- C) Smurf attack

D) Insider threat

Q11. What is the primary objective of a DoS attack?

A) Modify files

B) Steal passwords

C) Prevent legitimate users from accessing a service

D) Improve availability

Q12. What makes a DDoS attack different from a typical DoS attack?

A) It always targets wireless networks

B) It uses multiple systems to attack the target

C) It only attacks databases

D) It requires physical access

Q13. Which attack specifically abuses TCP connection establishment by sending large numbers of SYN requests?

A) Smurf attack

B) TCP SYN Flood

C) Session Hijacking

D) VLAN Hopping

Q14. Which attack is associated with an ICMP flood?

A) Smurf attack

B) Ransomware

C) Logical bomb

D) Phishing

Q15. A malicious employee intentionally uses their legitimate access to steal company information. What is this best classified as?

A) DDoS

B) Insider threat

C) Smurf attack

D) Physical environment attack

Q16. A malicious program is configured to execute when a particular condition or event occurs. Which attack is this?

A) Logical bomb

B) Phishing

C) Session hijacking

D) VLAN hopping

Q17. A user receives a fake message designed to trick them into revealing sensitive information. What attack is this?

A) Ransomware

B) Phishing

C) DoS

D) MITM

Q18. Which attack typically involves malicious software that prevents users from properly accessing their data or systems?

A) Ransomware

B) VLAN hopping

C) Smurf

D) Logical bomb

Q19. Which attack specifically attempts to move between VLANs in a way that violates the intended network segmentation?

A) TCP SYN Flood

B) VLAN hopping

C) Phishing

D) DDoS

Q20. Which of the following is a method of protecting a network mentioned in the chapter?

A) Removing all passwords

B) User training

C) Disabling every router

D) Increasing broadcast traffic

Q21. Scenario: A company wants to discover which ports are open on a target and gather information about the operating system. Which tool from the chapter is most closely associated with this task?

- A) Nessus
- B) Zenmap
- C) AUP
- D) Honey Net

Q22. Scenario: An administrator wants to scan a host and determine whether it is patched and whether it has received recent updates. Which tool should be considered?

- A) Zenmap
- B) Nessus
- C) Hashing
- D) MFA

Q23. What is the purpose of patching from a network-security perspective?

- A) To increase the number of vulnerabilities
- B) To address security weaknesses through updates
- C) To disable authentication
- D) To replace encryption with hashing

Q24. What is an Acceptable Use Policy (AUP) primarily intended to define?

- A) The physical layout of a network
- B) Which activities employees are allowed or prohibited from performing
- C) How routers calculate routes
- D) How encryption keys are mathematically generated

Q25. Scenario: A company policy specifies that employees cannot access other employees' personal files and that certain devices cannot be used at work. Which type of policy is this?

- A) AUP
- B) Routing policy
- C) Backup policy
- D) VLAN policy

Q26. Which part of a security policy specifies an approved standard, such as an approved type of encryption?

- A) Procedures
- B) Guidelines
- C) Standards
- D) Incident response

Q27. Which part of a security policy describes specific steps, such as how to create a new user or change a password?

- A) Standards
- B) Procedures
- C) Guidelines
- D) AUP

Q28. Scenario: A company requires an employee to enter a password and then use a security token to prove their identity. Which MFA factors are being combined?

- A) Something You Are + Somewhere You Are
- B) Something You Know + Something You Have
- C) Something You Do + Something You Are
- D) Something You Have + Somewhere You Are

Q29. Which MFA factor is represented by a fingerprint?

- A) Something You Know
- B) Something You Have
- C) Something You Are
- D) Something You Do

Q30. Scenario: A company's security team notices that users can still access the system, but some files are being secretly modified without authorization. Which CIA principle is being violated, and which security mechanism from the chapter is most directly relevant?

- A) Availability + DoS
- B) Confidentiality + Encryption

C) Integrity + Hashing

D) Availability + MFA

4. Answer Key

1-A | 2-B | 3-C | 4-C | 5-A

6-C | 7-B | 8-A | 9-C | 10-A

11-C | 12-B | 13-B | 14-A | 15-B

16-A | 17-B | 18-A | 19-B | 20-B

21-B | 22-B | 23-B | 24-B | 25-A

26-C | 27-B | 28-B | 29-C | 30-C

5. Final Memory Card

- CIA: Confidentiality = unauthorized viewing; Integrity = unauthorized modification; Availability = access to systems/data.
- Symmetric = same key; Asymmetric = pair of keys; Hashing = Integrity.
- MITM = attacker in the middle; Session Hijacking = takes over an established session.
- DoS/DDoS = Availability attacks; TCP SYN Flood = SYN requests; Smurf = ICMP flood.
- Phishing = deceptive message; Ransomware = malicious software affecting access to data/systems.
- AUP = allowed/prohibited use; Standards = approved requirements; Procedures = steps.
- MFA: Know = password; Have = token/phone; Are = fingerprint.